AP CALCULUS AB

Unit 1 • Lesson 1-9 • Guided notes

Lesson 1-9 • IVT, EVT, and MVT

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Choose the right existence theorem for a question.
- Check the hypotheses of the IVT, EVT and MVT from formulas, tables and graphs.
- Write complete AP-style justifications.

Vocabulary. existence theorem, hypothesis, conclusion, justification.

1. Which theorem?

Key idea

	Hypothesis	Conclusion
IVT	_____	$f(c) = k$ for every k between _____
EVT	_____	f has an absolute _____ on $[a,b]$
MVT	continuous on $[a,b]$ and _____	$f'(c) =$ _____ for some c in (a,b)

Questions about a _____ point to the IVT; about the _____, the EVT; about a _____, the MVT.

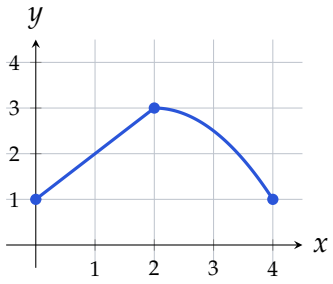
Example 1 • A table (AP style)

A differentiable function f has the values below.

x	0	2	5	7	10
$f(x)$	4	-2	3	9	5

(a) Must $f(c) = 0$ for some c in $[0,10]$? (b) Must $f'(c) = 3$ for some c in $(5,7)$? (c) Must f have an absolute maximum on $[0,10]$? (d) Must $f'(c) = 0$ for some c in $(0,10)$?

Example 2 • Hypotheses fail



The graph of f on $[0, 4]$ is shown (a corner at $x = 2$). Which theorems apply on $[0, 4]$?

AP exam tip
Never apply a theorem without stating why its hypotheses hold. On tables, "f is differentiable, so it is continuous" covers both continuity and differentiability.

Check your understanding

1. "Is there a time when the temperature was exactly $70°$?" uses the _____.
2. "Is there a time when the temperature was rising at $2°$ per hour?" uses the _____.